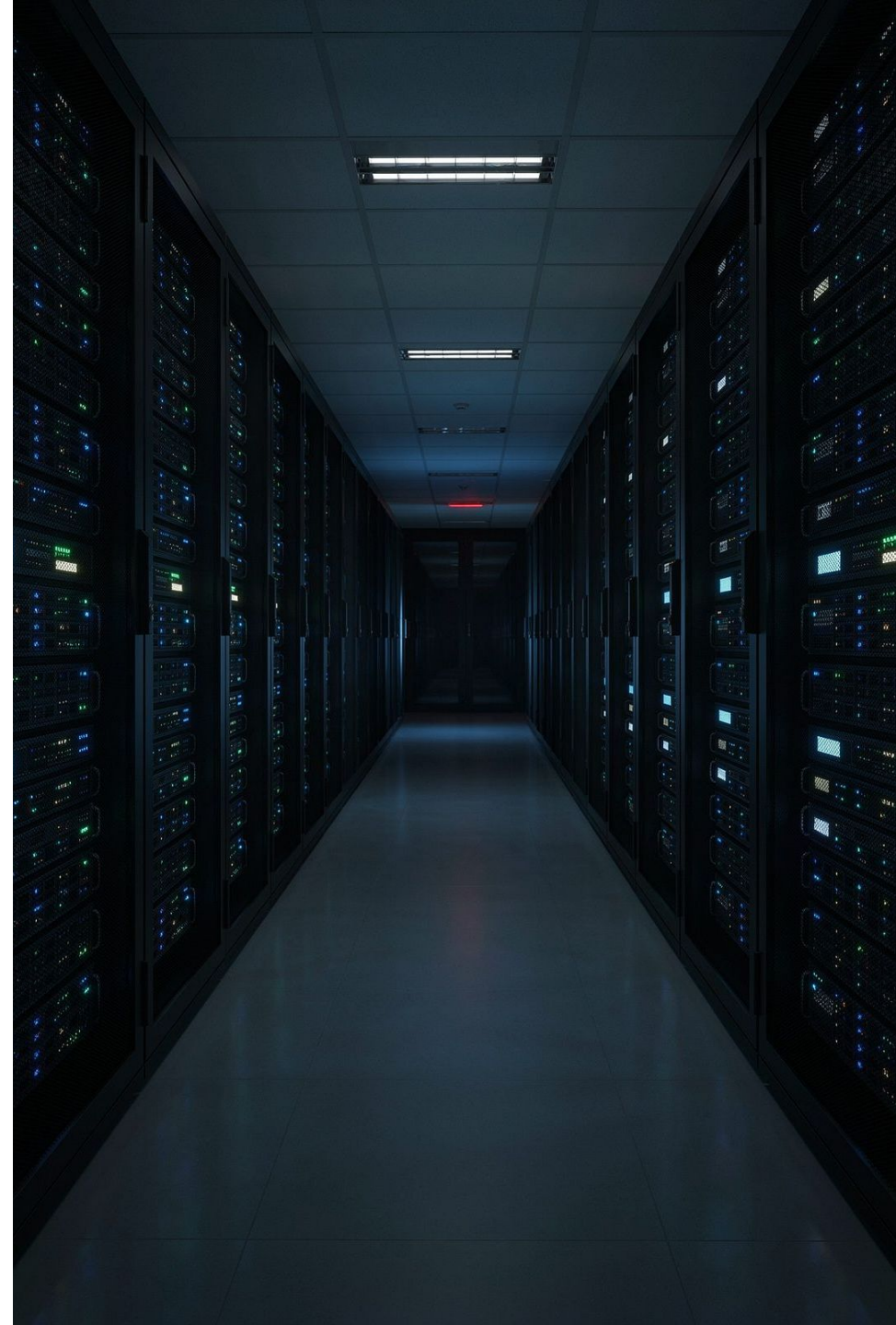


Windows & Linux Fundamentals

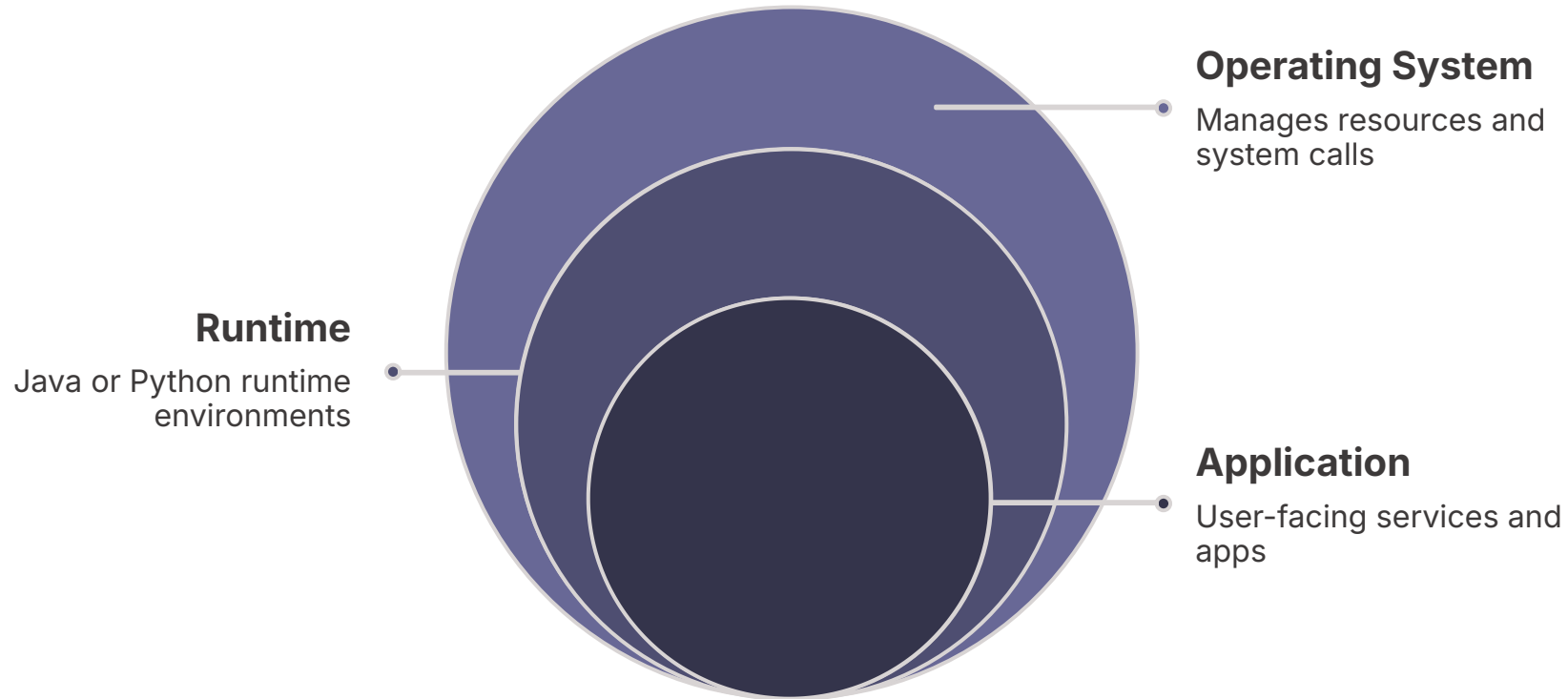
A practical guide to operating systems, shells, permissions, and secure developer practices for enterprise environments.

OS BASICS · SHELL · SECURITY



What Is an Operating System?

An OS sits between your application and the hardware. When Java needs to create a file, open a network connection, or allocate memory — it asks the OS, not the hardware directly.



In enterprise environments, developers work across both Windows (PowerShell) and Linux (Bash) — often daily.



Windows Fundamentals

Windows organizes storage using **drive letters**. Every path starts with a letter identifying the volume.

Common Drives

C: D: E:\

Developer Paths

C:\Users\Varun

C:\Projects\mortgage-
platform

System Directories

C:\Windows

C:\Program Files

C:\Users\<username>\AppData

Linux Fundamentals

Linux uses **no drive letters**. Everything starts from a single root: `/`. Each directory has a defined purpose.

Directory	Purpose
<code>/home</code>	User home directories
<code>/etc</code>	System configuration
<code>/var</code>	Logs and application data
<code>/tmp</code>	Temporary files
<code>/usr</code>	Programs and libraries
<code>/opt</code>	Optional software
<code>/bin</code>	Essential commands



Windows vs. Linux File Paths

Path syntax differs fundamentally — and mixing them up causes real bugs in cross-platform projects.

Windows

```
C:\Projects\mortgage-platform
```

Configuring tools matters:

- Java, Python, Maven
- Docker, Git
- CI/CD pipelines & scripts

Linux

```
/home/varun/projects/mortgage-platform
```



A script with `C:\data\borrowers.csv` will fail unchanged on Linux — and vice versa.

A photograph of two cardboard folders on a wooden surface. The folder on the left is labeled 'PROJECTS' and the one on the right is labeled 'ARCHIVE'.

Case Sensitivity: A Hidden Deployment Trap

Windows filesystems are typically **case-insensitive**. Linux treats filenames as **strictly case-sensitive**.

Windows — All the Same

`Borrower.csv`, `borrower.csv`, and `BORROWER.CSV` all resolve to the same file.

Linux — All Different

`Borrower.csv`, `borrower.csv`, and `BORROWER.CSV` are three distinct files.

Real Deployment Bug

Java expects `templates/BorrowerReport.html`, but the repo has `templates/borrowerReport.html`. Works on Windows. Fails on Linux.

Shell Fundamentals

A **shell** is your command-line interface to the OS. Today we work with both.

PowerShell (Windows)

Command	Action
Get-Location / pwd	Current directory
Get-ChildItem / ls	List files
cd C:\Projects	Change directory
mkdir mortgage-platform	Create directory
New-Item README.md	Create file
Get-Content README.md	Display file
Remove-Item README.md	Delete file

Bash (Linux / WSL)

Command	Action
pwd	Current directory
ls -la	List all (incl. hidden)
cd projects	Change directory
mkdir projects	Create directory
touch test.txt	Create file
cat archive.txt	Display file
rm test-copy.txt	Delete file

Pipelines, Variables & Scripts

PowerShell Pipeline

PowerShell pipes **objects**, not just text — making it powerful for automation.

```
Get-Process | Where-Object CPU -gt 10  
Get-Process | Sort-Object CPU -Descending | Select-Object -First 5
```

PowerShell Variables

```
$environment = "development"  
Write-Host "Starting in $environment"
```

Script: check-environment.ps1

```
java --version; python --version  
git --version; docker --version
```

Bash Variables

```
name="Mortgage Platform"  
echo $name  
export APP_ENV=development  
echo $APP_ENV
```

Script: check-environment.sh

```
#!/bin/bash  
java --version  
python3 --version  
git --version  
docker --version
```

```
chmod +x check-environment.sh  
./check-environment.sh
```

❏ Both scripts perform the same task — compare the syntax across shells.

Linux Permissions & Ownership

Every Linux file has an owner, a group, and three permission tiers.
Run `ls -l` to inspect them.

```
-rw-r--r-- 1 varun developers 1024 borrower.csv
```

Value	Bits	Meaning
7	rwX	Read, write, execute
6	rw-	Read and write
5	r-X	Read and execute
4	r--	Read only

```
chmod 755 script.sh # Owner=rwx, Group/Others=r-x
chmod 600 id_rsa    # Owner=rw, Group/Others=none
```

Ownership Commands

```
sudo chown varun borrower.csv
sudo chgrp developers borrower.csv
sudo chown varun:developers borrower.csv
```

⚠️ `chmod 600` on private keys is critical. Overly permissive credentials are a common security misconfiguration.

Three Permission Tiers

Owner

Group

Others

Process Management & Services

Windows — PowerShell

```
Get-Process  
Get-Process java  
Stop-Process -Name java  
Stop-Process -Id 1234
```

Windows — Services

```
Get-Service | Where-Object {$_.Name -like "*postgres*"}  
Start-Service postgresql-x64-17  
Stop-Service postgresql-x64-17
```

Linux — Bash

```
ps aux | grep java  
top  
kill PID      # graceful  
kill -9 PID   # force — use sparingly
```

Linux — systemd Services

```
systemctl status postgresql  
sudo systemctl start postgresql  
sudo systemctl restart postgresql
```

 `kill -9` forces termination without cleanup — never make it your first troubleshooting step.



Enterprise Workstation Hardening

A corporate developer workstation is a **controlled endpoint** connected to sensitive enterprise systems — not just a coding machine.



Identity & Access

Enable MFA. Use company-approved VPN. Use separate developer credentials. Avoid admin privileges for routine work.



Secrets Hygiene

Never store credentials in source code. Never commit `.env` files. Use SSH keys or credential managers.



Endpoint Hygiene

Apply Windows security updates. Keep endpoint protection active. Lock workstation when unattended. Use approved software repositories only.

PII & Mortgage Data Handling

PII (Personally Identifiable Information) in a mortgage context includes borrower names, addresses, SSNs, bank details, income, credit data, and loan documents.

⊗ If someone sends you `production_borrower_data.csv` — do not copy it to Downloads, Desktop, personal cloud storage, or USB drives.

Enterprise policy typically requires masked or synthetic data in development, controlled encrypted storage, restricted folder access, and full audit logging.

